Data Quality Report

Executive summary

Every one of the seven forensic checks the assignment asked for (A–G) returned a real, quantified answer — four confirmed problems, three ruled out. The two confirmed identity/attribution problems (borrower_id corruption, agent identity collapse) are severe enough that they redefine which fields in this dataset can be trusted at all, and that decision governs every downstream number in this submission.

Issues found, detection method, treatment, business impact

#	Issue	Detection	Treatment	Business impact
1	`borrower_id` wrong in ~98% of rows across **every** event table (calls, payments, PTPs, field visits, complaints, dispositions, call_attempts, WhatsApp, SMS)	Joined each event table to `accounts.csv` (the one clean 1:1 dimension) on `account_id` and compared `borrower_id` values — mismatch rate 96.9%–98.5% depending on table	Dropped `borrower_id` from every event table; re-derived via `account_id` → `accounts.borrower_id` at the golden layer	Any prior analysis joining on event-table `borrower_id` (e.g. "borrower-level" recovery or complaint history) was silently wrong for ~98% of rows

2	agents.csv: 30,000 rows but only 1,000 distinct agent_id; for a given agent, name/team/vendor/status/employee_code are randomized across its ~30 rows, not a real change history	Grouped by agent_id, inspected all rows for one agent — attributes showed no temporal or logical pattern; employee_code maps to 40+ different agent_ids	agent_id kept as the only trustworthy key; team/vendor derived by mode (most frequent value) per agent with an explicit attribute_confidence flag (HIGH only if all 30 rows agreed, else LOW)	Any agent-team or agent-vendor performance comparison must carry a confidence caveat; employee_code cannot be used as an identity key at all
3	borrowers.csv: 30,600 rows but only 11,015 distinct borrower_id; same randomization pattern as agents (name/phone/email/city/state scrambled across duplicates)	Same grouping/inspection method as #2	Deduplicated to one row per borrower_id via deterministic tie-break (latest created_at); flagged low-confidence	**Geography and borrower-demographic analysis is fundamentally unreliable** from this file — stated explicitly rather than silently used
4	Duplicate payments, two mechanisms: (a) 500 exact duplicate payment_id rows; (b) retry chains under shared payment_reference with churning status (FAILED→PENDING→SUCCESS→REVERSED)	Compared COUNT(*) vs COUNT(DISTINCT payment_id); grouped by payment_reference and inspected status arrays per group	Deduped by payment_id; collapsed each payment_reference chain to its final-by-timestamp state; excluded chains that were ever REVERSED even if SUCCESS appeared earlier	Naive "SUCCESS" sum overstated real recovery by 6%–29% depending on month — the single largest correction affecting the headline "11%" claim

5	Mixed timezones (UTC / Asia/Kolkata / Asia/Dubai) on every event table with an `event_at`	Distinct-value scan of each table's `timezone` column	Normalized every `event_at` to IST using the row's own `timezone` label before any hour-of-day or day-of-week analysis	Calling-time-of-day analysis would have been meaningless /wrong without this — up to 5.5 hours of misclassification per row
6	Exact duplicate `call_id` rows (91,350 → 90,000 distinct)	`COUNT(*)` vs `COUNT(DISTINCT call_id)`	Deduped, keep earliest `event_at`	Call-volume-based metrics (contact rate, attempts) would be inflated ~1.5% if left uncorrected
7	`disposition_code` vocabulary changed across `disposition_version`: legacy scheme has both `PROMISE_TO_PAY` and `PTP` as separate codes; v1/v2 use only PTP	Grouped `disposition_code` by `disposition_version`, compared code sets	Normalized `PROMISE_TO_PAY` → `PTP`	Prevents a spurious PTP-rate "trend" that is really just a code-naming artifact
8	`field_visits.outcome` values assumed incorrectly on first pass (checked for a non-existent `'NOT_HOME'` value)	Caught during QA of `golden_interactions` — field-channel contact rate came back as a suspicious flat 100%	Corrected to the real value set: `PAID/CONTACTED/REFUSED/PTP` = contacted, `NOT_AVAILABLE/WRONG_ADDRESS` = not contacted	Self-caught before submission; documented here for transparency about the QA process itself

Checks that came back clean (equally important to report)

Check	Method	Result
Vendor telephony mapping changes (Part 2D)	Monthly call volume per `vendor_id`	Stable, near-uniform volume every month across all 15 vendors — **ruled out** as a driver
Portfolio mix changes (Part 2F)	`risk_segment` share of accounts, both at origination and in the monthly worked population	Stable ~25% per segment every month — **ruled out**
Denominator manipulation (Part 2G)	`daily_targeting.status` share (QUEUED/CONTACTED/SKIPPED/EXPIRED) by month	Flat ~25% each, every month — no accounts quietly disappearing from the denominator — **ruled out**

Cleaning-decision impact, quantified (Raw → Rejected/Corrected → Golden)

Table	Raw rows	Rejected/corrected	Golden rows	Correction
accounts	30,000	0	30,000	None needed — clean 1:1 dimension
borrowers	30,600	19,585 collapsed	11,015	Deduped to distinct `borrower_id`, low-confidence attributes flagged
agents	30,000	29,000 collapsed	1,000	Deduped to distinct `agent_id`, attributes derived by mode
calls	91,350	1,350 removed	90,000	Exact-duplicate `call_id` removal

payments	25,500	4,310 removed/collapsed	21,190	500 exact dupes + retry-chain collapse to final state
Recovery amount (Jan–Jul)	**₹126.3 Cr (naive SUCCESS sum)**	**–₹19.3 Cr**	**₹107.0 Cr**	**15.3% overstatement corrected**

Assumptions carried forward

- Timezone offsets applied as fixed IST conversions (UTC +5:30, Dubai +1:30); does not account for any DST-equivalent shifts (India and UAE do not observe DST, so this is safe).
- Payment-reference retry-chain resolution assumes the last-by-timestamp status is the authoritative final state; this is the standard payment-gateway reconciliation convention but is an assumption, not a documented business rule (none was provided).
- PTP-kept-rate uses a 14-day post-promise window based on typical collections-industry practice — the initial 3-day window was tested and rejected (see analysis log) because it produced an implausible <1% kept rate against a 56-day median PTP-to-payment gap in the data.
- No cost/pricing fields exist anywhere in the raw dataset; all unit-economics figures used in the ₹10 Cr investment recommendation are explicitly labeled assumptions, not derived from data.